



WORK SHEET - 11

ELLIPSE & HYPERBOLA

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.8 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 If the line $y = ax + b$ is tangent to the curve $f(x) = x - x^3$ at the point $(-1, 0)$ then the eccentricity of the ellipse $\frac{x^2}{a^4} + \frac{y^2}{b^2} = 1$ is
- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{2}{3}$
- Q.2 Let LL' be the latus rectum through the focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and A' be the farther vertex. If $\Delta A'LL'$ is equilateral, then the eccentricity of the hyperbola is
- (A) $\sqrt{3}$ (B) $\sqrt{3} + 1$ (C) $\frac{\sqrt{3} + 1}{\sqrt{2}}$ (D) $\frac{\sqrt{3} + 1}{\sqrt{3}}$
- Q.3 The curve $y = (|x| - 1) \operatorname{sgn}(x - 1)$ divides $\frac{9x^2}{64} + \frac{4y^2}{25} = \frac{1}{\pi}$ into two parts of areas A_1 and A_2 ($A_1 < A_2$), then
[Note : $\operatorname{sgn}(k)$ denotes signum function of k .]
- (A) $A_1 = \frac{7}{6}$ (B) $A_1 : A_2 = 7 : 13$ (C) $A_2 = \frac{13}{6}$ (D) $A_2 - A_1 = 1$
- Q.4 The locus of centre of circle touching the circles $(x + 3)^2 + y^2 = 16$ and $(x - 3)^2 + y^2 = 4$ externally is
- (A) $\frac{x^2}{8} - y^2 = 1$ (B) $x^2 - \frac{y^2}{8} = 1$ (C) $\frac{x^2}{4} - y^2 = 1$ (D) $x^2 - \frac{y^2}{4} = 1$
- Q.5 If normal at any point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$) meet the major and minor axes at Q and R respectively so that $3PQ = 2PR$, then the eccentricity of ellipse is equal to
- (A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{2}}$
- Q.6 If the ellipse $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{9} - \frac{y^2}{7} = 1$ intersects orthogonally, then b is equal to
- (A) $\sqrt{3}$ (B) 3 (C) $2\sqrt{3}$ (D) 9
- Q.7 The point on the ellipse $x^2 + 2y^2 = 6$ which is nearest to the line $x - y = 7$ is
- (A) (2, 1) (B) (2, -1) (C) (-2, 1) (D) (-2, -1)



- Q.8 If the line $\frac{x}{a} + \frac{y}{b} = \sqrt{2}$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$), then the eccentric angle of the point of contact is
- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$

[PARAGRAPH TYPE]

Q.9 to Q.11 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 9 to 11

Let the line $2mx + y\sqrt{1-m^2} = 1$ touches a fixed ellipse whose axes are coordinate axes, for infinitely many real values of m .

- Q.9 The equation of auxiliary circle of the ellipse is
- (A) $x^2 + y^2 = \frac{1}{4}$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = \frac{3}{4}$ (D) $x^2 + y^2 = \frac{1}{2}$
- Q.10 The foci of ellipse are
- (A) $(0, \pm\sqrt{3})$ (B) $(\pm\sqrt{3}, 0)$ (C) $\left(\pm\frac{\sqrt{3}}{2}, 0\right)$ (D) $\left(0, \pm\frac{\sqrt{3}}{2}\right)$
- Q.11 The locus of point of intersection of perpendicular tangents of the ellipse is
- (A) $x^2 + y^2 = 2$ (B) $x^2 + y^2 = \frac{3}{2}$ (C) $x^2 + y^2 = \frac{5}{4}$ (D) $x^2 + y^2 = \frac{7}{4}$

[MULTIPLE CORRECT CHOICE TYPE]

Q.12 to Q.16 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.12 A normal to the hyperbola $x^2 - 4y^2 = 4$ meets the x and y axes at A and B respectively. The locus of point of intersection of the straight lines drawn through A and B perpendicular to the x and y axes respectively is a hyperbola. Then
- (A) eccentricity equals $\sqrt{3}$ (B) length of latus rectum equals 20.
 (C) equation of auxiliary circle equals $x^2 + y^2 = 25$. (D) distance between foci equals $5\sqrt{5}$
- Q.13 If a pair of variable straight lines $x^2 + 4y^2 + \alpha xy = 0$ (α is real parameter) cuts the ellipse $x^2 + 4y^2 = 4$ at two points A and B, then the locus of the point of intersection of tangents at A and B of ellipse is
- (A) $x - 2y = 0$ (B) $2x - y = 0$ (C) $x + 2y = 0$ (D) $2x + y = 0$
- Q.14 Let P be a point on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. The tangent drawn at P meets the coordinate axes at A and B, then
- (A) minimum area of ΔOAB is 6. (B) minimum area of ΔOAB is 12.
 (C) minimum length of AB is 5. (D) minimum length of AB is $\sqrt{13}$.
 (where O is origin)

Q.15 Tangents are drawn from points on the line $3x + 4y = 24$ to the curve $x^2 + \frac{y^2}{4} = 1$.

The point of concurrency of the chords of contact lie on the curve

- (A) $16x - 3y = 0$ (B) $9y^2 = 32x$ (C) $24x + 24y = 19$ (D) $y = 2x$

Q.16 Two lines are drawn from point $P(\alpha, \beta)$ which touches $y^2 = 8x$ at A, B and touches $\frac{x^2}{4} + \frac{y^2}{6} = 1$ at

C, D, then

- (A) $\alpha + \beta = -4$ (B) $\alpha\beta = 4$
(C) Area of $\Delta PAB = 128\sqrt{2}$ (D) Area of $\Delta PAB = 32\sqrt{2}$

[INTEGER TYPE]

Q.17 to Q.18 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.17 If the locus of centres of the circles which cut the ellipse $\frac{x^2}{8} + \frac{y^2}{6} = 1$, at $P(2, \sqrt{3})$ orthogonally is

$\sqrt{a}x + 2y = b\sqrt{c}$, where $a, b, c \in \mathbb{N}$ and b, c are relatively prime, then find the value of $\left\lfloor \frac{a+b+c}{2} \right\rfloor$.

Q.18 Consider an ellipse $E: \frac{(3x-4y+5)^2}{100} + \frac{(4x+3y-10)^2}{50} = 1$. M and m are lengths of major and minor

axes respectively. If p_1 and p_2 are perpendicular distances on major and minor axes from a point P in the x - y plane such that $m \leq p_1 + p_2 \leq M$, then P lies in the region whose area is Δ . Find the sum of the digits in Δ .

ANSWER KEY

Q.1	C	Q.2	D	Q.3	B	Q.4	B	Q.5	A
Q.6	B	Q.7	B	Q.8	A	Q.9	B	Q.10	D
Q.11	C	Q.12	BD	Q.13	AC	Q.14	AC	Q.15	ABC
Q.16	AD	Q.17	[5]	Q.18	[7]				





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